

PROVISIONAL PATENT APPLICATION

REMOVABLE PASSIVE RESERVOIR INSERT FOR A PLANTER

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TITLE OF THE INVENTION

Removable Passive Reservoir Insert for a Planter

FIELD OF THE INVENTION

The present disclosure relates to horticultural containers and, more particularly, to a removable support insert that creates a water-storage region below growing medium in an existing planter.

BACKGROUND OF THE INVENTION

Most conventional planters depend on repeated watering and drainage through openings in the bottom of the container. Water may drain too quickly, collect unevenly, or remain in direct contact with the root zone. Fast drainage increases watering frequency. Standing water around the roots can contribute to poor plant health.

Self-watering planters and reservoir systems are available, but many use a dedicated planter body, several fitted parts, powered equipment, or a permanent internal structure. A user who already owns a planter may have to replace it to obtain those features. There is therefore a need for a simple removable insert that creates a passive reservoir in a conventional planter and supports the growing medium above the stored water.

SUMMARY OF THE INVENTION

The invention is a removable planter insert with an outer support portion and an open interior support structure. The insert is placed above the bottom of a planter. It supports growing medium and a plant root zone while leaving a water-storage region below. Openings in the structure allow water to drain into the lower region and allow moisture to move back toward the growing medium.

In one embodiment, the interior support structure includes spaced concentric rings connected by intersecting braces. This arrangement spreads the supported load while leaving open flow paths and using relatively little material. The insert may be additively manufactured as one piece. It may be removed for cleaning, reuse, replacement, or transfer to another compatible planter.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a top engineering view of a Revision 5 planter insert having an overall diameter of approximately 4.25 inches.

FIG. 2 is a side engineering view of the Revision 5 planter insert.

FIG. 3 is a multi-view engineering drawing of an earlier Iteration 1.4 planter insert.

FIG. 4 is a rendered perspective view of the planter insert.

FIG. 5 is a static stress simulation image for an example design iteration.

DETAILED DESCRIPTION OF THE INVENTION

A removable planter insert includes a generally circular outer rim surrounding an open, load-supporting interior. In the illustrated embodiment, spaced concentric rings form much of the interior. Two braces cross near the center and connect the rings to the outer rim. Open spaces remain between the rings and braces.

The insert is sized to fit in the lower part of a conventional planter. Revision 5 has an overall diameter of approximately 4.25 inches and was developed for an approximately six-inch planter. Those dimensions describe one example and are not limiting. The insert may be scaled or modified for different planter diameters, tapers, bottom contours, wall thicknesses, and desired reservoir depths.

During installation, the insert is positioned above the planter bottom. The rim and interior structure support growing medium placed over the insert. The space between the insert and the bottom of the planter becomes a reservoir. Water added from above passes through the growing medium and the openings in the insert. Moisture can then move upward through contact with the growing medium, through a wick or other capillary element, or through limited root growth into the openings.

The insert keeps most of the supported root zone out of continuous contact with water at the bottom of the planter. The openings still allow drainage, airflow, and moisture movement. Because the insert is not permanently attached to the planter, a user may remove it without damaging the planter. The insert can be cleaned or replaced separately.

The insert may be additively manufactured from a suitable polymer. Other versions may be molded, machined, cast, assembled from several parts, or made from another water-compatible material. Ring and brace dimensions may be selected for the material, expected load, manufacturing process, and desired open area.

The concentric-ring arrangement is one example. Other versions may use a lattice, perforated plate, radial or curved ribs, polygonal cells, or a combination of structures. The outer support may have a different shape, and feet, spacers, or a ledge may set the reservoir height.

Optional features may include a wick, fill tube, overflow passage, drain opening, filter, water-level indicator, or sensor. These features are not required for the basic passive insert to work.

A static stress study was performed on an example CAD model. The software displayed a safety factor of 15 for the stated model and load case. That result suggests the evaluated geometry may allow material reduction, but it is not a load rating for every material, print condition, planter, or use. Physical testing is still needed for a final manufactured design.

EXAMPLE METHOD OF USE

A user selects an insert sized for the planter and places it above the planter bottom. The user adds growing medium over the insert, places a plant in the growing medium, and adds water. Some of the water passes through the growing medium and the open insert into the lower reservoir. The insert supports the growing medium above the reservoir while allowing passive moisture transfer.

CLAIMS

1. A removable insert for a planter, comprising: an outer support portion configured to be received within the planter; and an open interior support structure connected to the outer support portion and configured to support growing medium above a bottom region of the planter, thereby defining a water-storage region below the growing medium while permitting water movement through the open interior support structure.
2. The removable insert of claim 1, wherein the open interior support structure comprises a plurality of spaced concentric rings.
3. The removable insert of claim 2, further comprising intersecting braces connecting the plurality of spaced concentric rings to the outer support portion.
4. The removable insert of claim 1, wherein the outer support portion and the open interior support structure form a unitary additively manufactured component.
5. The removable insert of claim 1, wherein the insert is removable from the planter for cleaning, reuse, replacement, or transfer to another planter.
6. The removable insert of claim 1, further comprising a capillary element configured to transfer water from the water-storage region toward the growing medium.
7. The removable insert of claim 1, further comprising at least one of a fill passage, an overflow passage, a drain passage, or a water-level indicator.
8. The removable insert of claim 1, further comprising a water-level sensor or a soil-moisture sensor.

9. A planter system comprising: a planter having a bottom region; the removable insert of claim 1 positioned above the bottom region; growing medium supported above the removable insert; and a water-storage region located between the removable insert and the bottom region.
10. A method of forming a passive reservoir in a planter, comprising: positioning a removable open support insert above a bottom of the planter; supporting growing medium above the insert; and receiving water through the growing medium and the insert within a reservoir region located below the insert.

ABSTRACT

A removable passive reservoir insert is configured for placement in an existing planter. The insert includes an outer support portion and an open interior support structure that supports growing medium and a plant root zone above a lower water-storage region. In one embodiment, spaced concentric rings are connected by intersecting braces. Openings permit drainage and passive moisture movement while keeping most of the supported root zone out of continuous contact with stored water. The insert may be additively manufactured as one piece and removed for cleaning, reuse, resizing, or replacement. Optional features may include capillary elements, fill or overflow passages, a water-level indicator, or sensing components.



FIG. 1 - Perspective Rendering

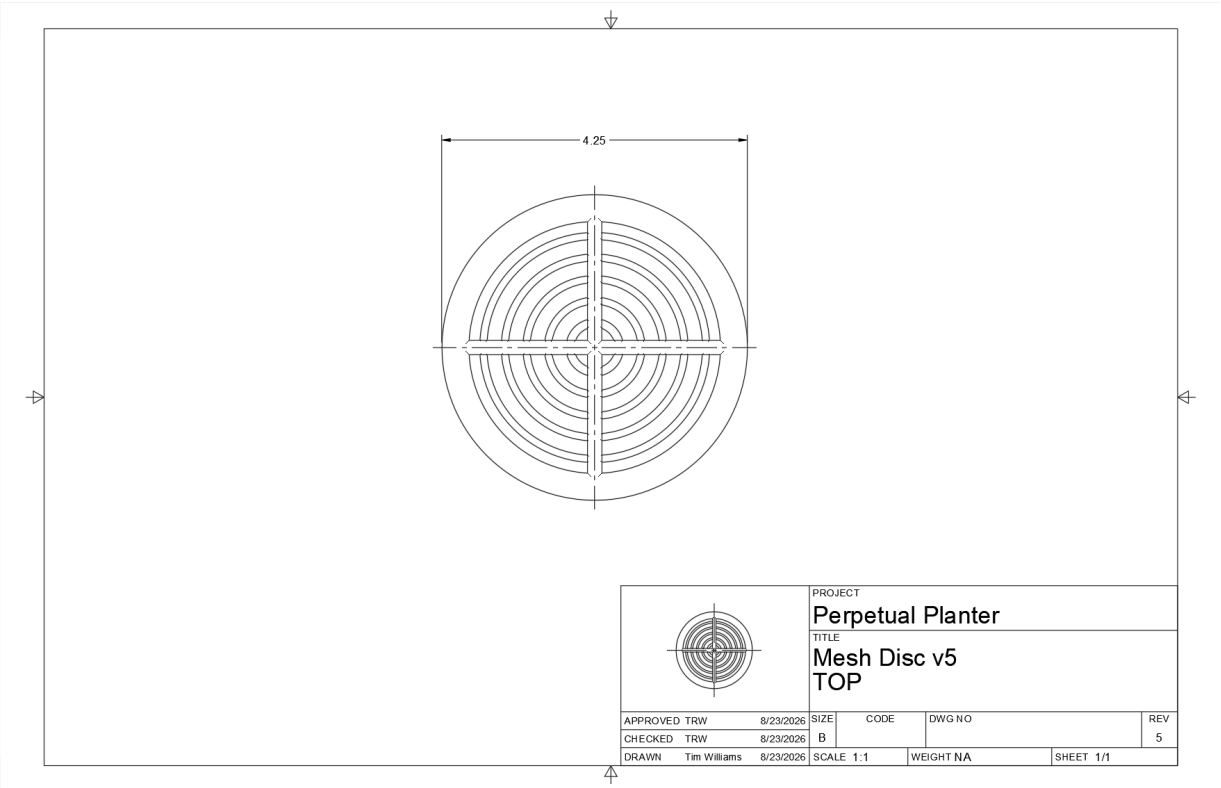


FIG. 2 - Revision 5 Top View

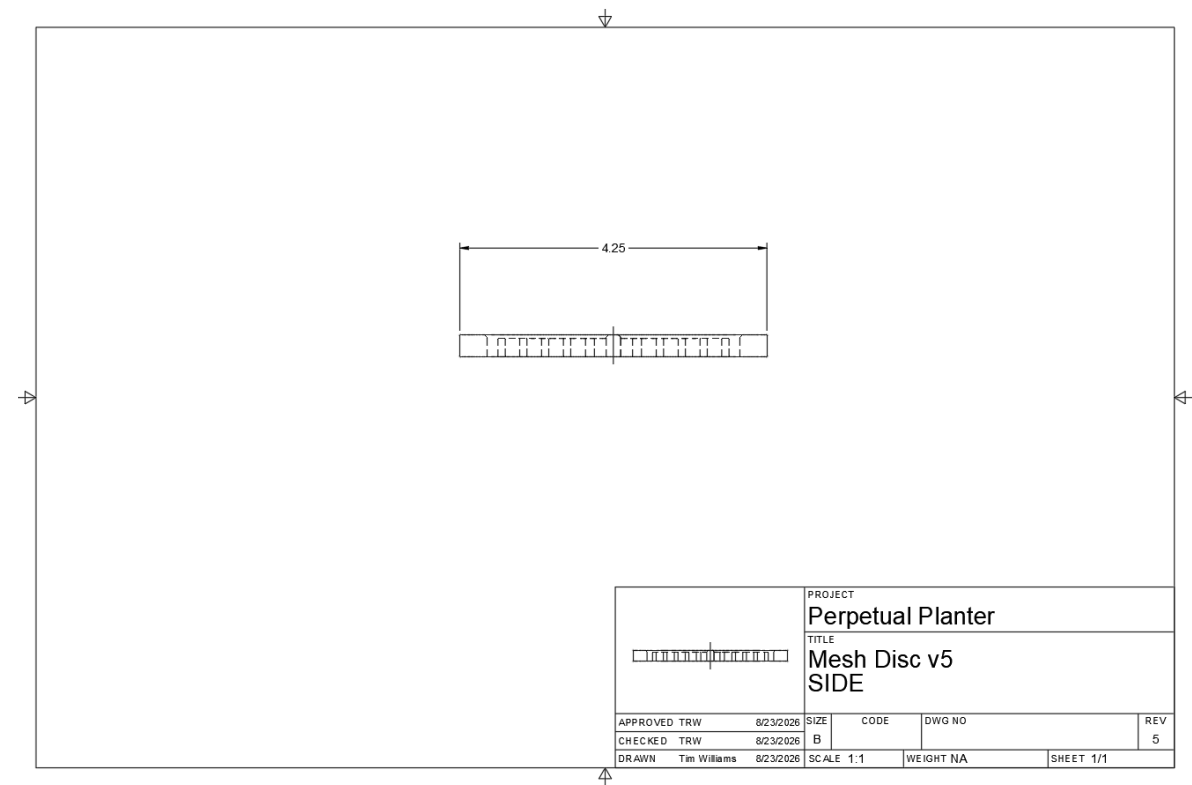


FIG. 3 - Revision 5 Side View

DRAWINGS:

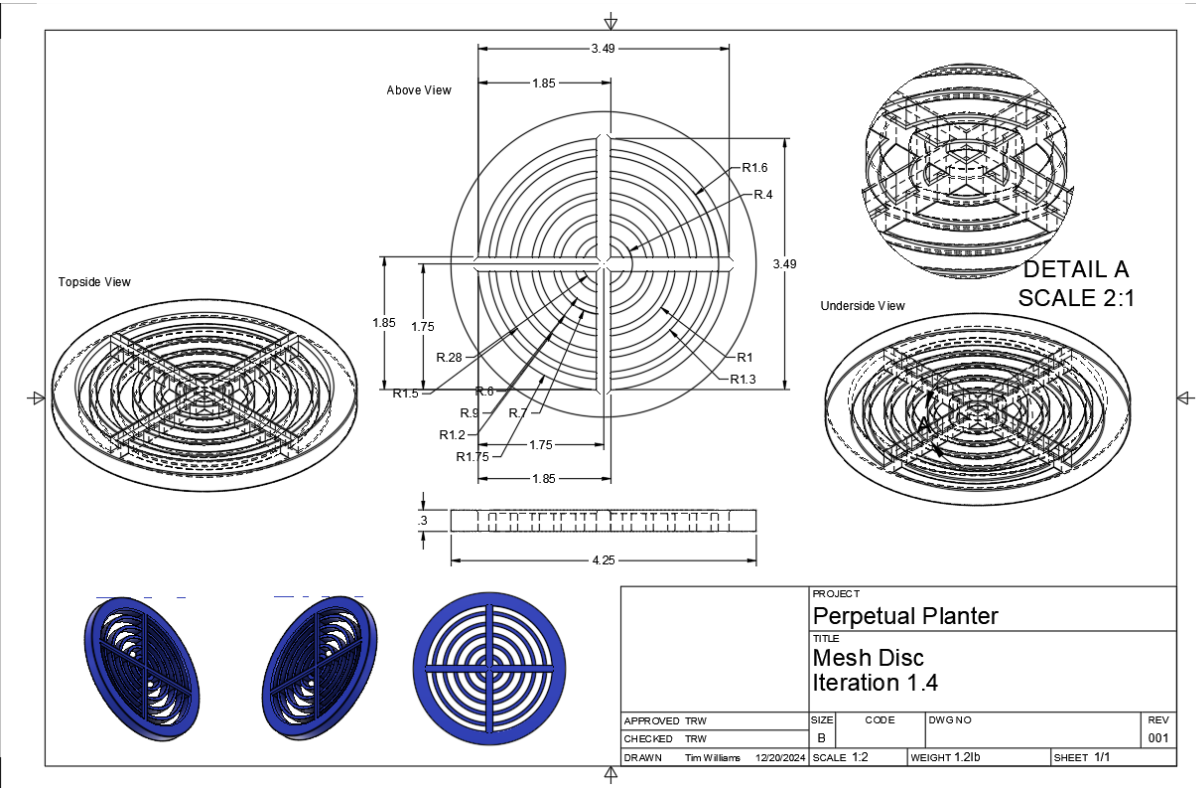
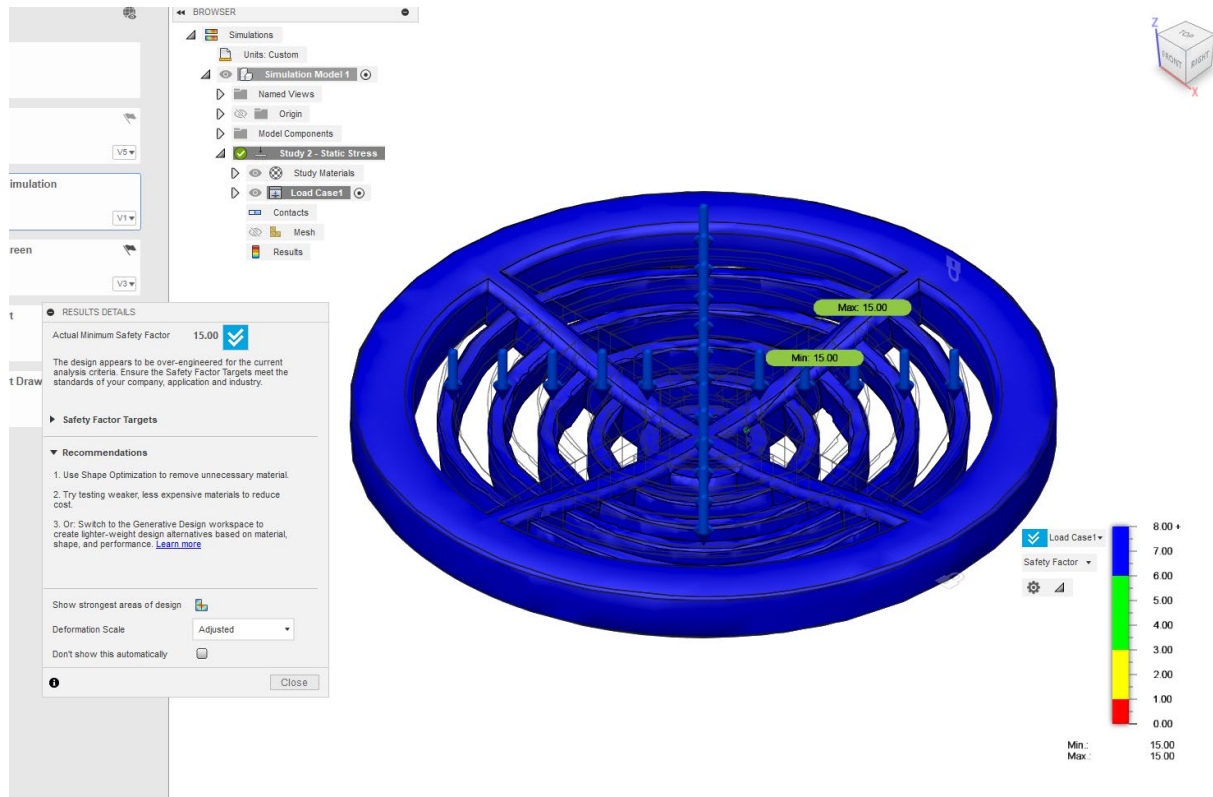


FIG. 4 - Iteration 1.4 Multi-View Drawing

DRAWINGS:**DRAWINGS:***FIG. 5 - Example Static Stress Study*